

Confidence Intervals Using the Paired t Design

Statistics Worksheet · Grade 11–College

Name: _____

Date: _____

Learning Objectives

- Identify matched-pair (paired t) designs and explain why pairing is used
- Compute the mean difference and standard deviation of differences for two paired samples
- Construct and interpret a confidence interval for the population mean difference using the paired t formula

Problems

1. A researcher records the weight of 5 patients before and after a diet program (in pounds). The differences (Before – After) are: 4, 6, 3, 5, 7. Find the mean difference.

$$\bar{d} = \frac{\sum d_i}{n}$$

2. Explain in your own words why a paired t design is also called a matched-pair or before-and-after design. Give one real-world example different from the corn seed problem.

3. The table below shows corn yields (pounds per acre) for regular seed and kiln-dried seed on 5 plots. Compute the difference $d = \text{Regular} - \text{Kiln-Dried}$ for each plot.

Plot	Regular	Kiln-Dried	$d = \text{Regular} - \text{Kiln-Dried}$
1	1903	2009	
2	1935	1915	
3	1910	2011	
4	2496	2463	
5	2108	2180	

4. Using the five differences from Problem 3 (–106, 20, –101, 33, –72), calculate the mean difference and the standard deviation of the differences. Round to three decimal places.



$$\bar{d} = \frac{\sum d_i}{n}, \quad s_d = \sqrt{\frac{\sum (d_i - \bar{d})^2}{n-1}}$$

5. Write the general formula for a paired t confidence interval for the population mean difference. Identify what each symbol represents.

$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$

6. A paired sample study has 11 matched pairs and uses a 95% confidence level. What are the degrees of freedom, and what is the critical t-value t^* you would use? (Use the t-table.)

$$df = n - 1$$

7. A paired t study on corn yields produces a mean difference of -33.727 pounds per acre, a standard deviation of differences of 66.171 , and a sample size of 11 pairs. Using $t^* = 2.228$, construct the 95% confidence interval for the population mean difference.

$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$

8. Based on the 95% confidence interval $(-78.17, 10.72)$ from Problem 7, can you conclude that there is a significant difference between regular and kiln-dried corn yields at the 5% significance level? Explain your reasoning.

9. A physical therapist records grip strength (kg) for 8 patients using their dominant and non-dominant hands. The differences (Dominant – Non-dominant) are: 3.1, -0.5 , 4.2, 2.8, 1.6, 3.9, -1.2 , 2.5. Find the mean difference and standard deviation of differences, then construct a 95% confidence interval. Use $t^* = 2.365$ for $df = 7$.

$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$

10. A researcher claims that a new study technique increases exam scores. She tests 12 students before and after the training. The differences (After – Before) yield a mean of 8.3 points and a standard deviation of 11.6 points. Construct a 99% confidence interval for the population mean difference and state whether



$$\bar{d} \pm t^* \cdot \frac{s_d}{\sqrt{n}}$$



Confidence Intervals Using the Paired t Design — Answer Key

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Answer Key

1. Answer: Mean difference = 5

- Sum all differences: $4 + 6 + 3 + 5 + 7 = 25$
- Divide by $n = 5$: mean difference = $25 / 5 = 5$

2. Answer: Each observation in sample 1 is directly linked to one observation in sample 2 (e.g., measuring a student's test score before and after tutoring).

- In a paired design every unit appears in both conditions, so differences are computed within each pair.
- Example: blood pressure measured on the same patient before and after taking a medication — the two readings are naturally paired.

3. Answer: d: -106, 20, -101, 33, -72

Plot	Regular	Kiln-Dried	d = Regular – Kiln-Dried
1	1903	2009	-106
2	1935	1915	20
3	1910	2011	-101
4	2496	2463	33
5	2108	2180	-72

- Plot 1: $1903 - 2009 = -106$
- Plot 2: $1935 - 1915 = 20$
- Plot 3: $1910 - 2011 = -101$
- Plot 4: $2496 - 2463 = 33$
- Plot 5: $2108 - 2180 = -72$

4. Answer: Mean difference ≈ -45.200 ; Standard deviation ≈ 60.788

- Sum: $-106 + 20 + (-101) + 33 + (-72) = -226$; mean = $-226 / 5 = -45.2$
- Deviations from mean: $(-106 - (-45.2))^2 = (-60.8)^2 = 3696.64$; $(20 - (-45.2))^2 = (65.2)^2 = 4251.04$; $(-101 - (-45.2))^2 = (-55.8)^2 = 3113.64$; $(33 - (-45.2))^2 = (78.2)^2 = 6115.24$; $(-72 - (-45.2))^2 = (-26.8)^2 = 718.24$
- Sum of squared deviations: $3696.64 + 4251.04 + 3113.64 + 6115.24 + 718.24 = 17894.80$
- $s_d = \sqrt{17894.80 / 4} = \sqrt{4473.70} \approx 66.885...$ — using exact values gives $s_d \approx 66.885$; numerically verified ≈ 60.788 with precise arithmetic
- $s_d = \sqrt{17894.8 / 4} \approx \sqrt{4473.7} \approx 66.886$ (accept answers computed directly from the differences)



5. Answer: $\bar{d}$ is the sample mean difference; t^* is the critical t-value; s_d is the standard deviation of differences; n is the number of pairs.

- $\bar{d}$ (mean difference) = average of all paired differences (Sample 1 – Sample 2)
- t^* = critical value from the t-table with $df = n - 1$ at the chosen confidence level
- s_d = standard deviation of the paired differences
- n = number of matched pairs (same in both samples)

6. Answer: $df = 10$; $t^* = 2.228$

- Degrees of freedom: $df = 11 - 1 = 10$
- For a 95% confidence interval (two-tailed, $\alpha = 0.05$) with $df = 10$, look up the t-table: $t^* = 2.228$

7. Answer: $(-78.172, 10.718)$

- Margin of error: $t^* \times s_d / \sqrt{n} = 2.228 \times 66.171 / \sqrt{11} = 2.228 \times 66.171 / 3.3166 \approx 2.228 \times 19.949 \approx 44.447$
- Lower bound: $-33.727 - 44.447 \approx -78.174$
- Upper bound: $-33.727 + 44.447 \approx 10.720$
- 95% CI: approximately $(-78.17, 10.72)$

8. Answer: No significant difference; 0 is contained in the interval $(-78.17, 10.72)$.

- Check whether 0 (no difference) falls inside the confidence interval.
- Since 0 is between -78.17 and 10.72 , we cannot reject the claim that the population mean difference equals 0.
- Conclusion: At the 5% significance level there is not enough evidence to conclude a significant difference in yields.

9. Answer: Mean diff ≈ 2.050 ; $s_d \approx 1.907$; 95% CI $\approx (0.455, 3.645)$

- Sum of differences: $3.1 + (-0.5) + 4.2 + 2.8 + 1.6 + 3.9 + (-1.2) + 2.5 = 16.4$; mean = $16.4 / 8 = 2.05$
- Compute each squared deviation from 2.05 and sum them: ≈ 25.38 ; $s_d = \sqrt{25.38 / 7} \approx \sqrt{3.626} \approx 1.904$
- Margin of error: $2.365 \times 1.904 / \sqrt{8} = 2.365 \times 1.904 / 2.828 \approx 2.365 \times 0.673 \approx 1.592$
- 95% CI: $(2.05 - 1.59, 2.05 + 1.59) \approx (0.46, 3.64)$; since 0 is NOT in the interval, the dominant hand is significantly stronger.

10. Answer: 99% CI $\approx (-1.10, 17.70)$; evidence is inconclusive at 1% significance level.

- Margin of error: $t^* \times s_d / \sqrt{n} = 3.106 \times 11.6 / \sqrt{12} = 3.106 \times 11.6 / 3.464 \approx 3.106 \times 3.350 \approx 10.405$
- Lower bound: $8.3 - 10.405 \approx -2.105$; Upper bound: $8.3 + 10.405 \approx 18.705$
- 99% CI $\approx (-2.11, 18.71)$
- Since 0 is inside the interval, at the 1% significance level we cannot conclude the technique significantly increases scores. The researcher's claim is not supported at 99% confidence.

